25900237

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular Examination October 2025
THIRD SEMESTER
Discipline Specific Core Course (DSC) – MG3DSCBIF201-APPLIED MATHEMATICS

(2024 ADMISSION ONWARDS)

Part A

Short Answer Questions (2 marks each)
Answer any 5 questions
(Total = 5 × 2 = 10 marks)

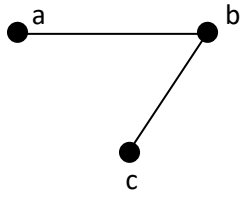
1. Define a singleton set with an example.
   - A set containing exactly one element. **1 mark**

   Example: $A = \{5\}$ or $B = \{a\}$. **1 mark**
2. Which law is applied in $(Y \vee F) \equiv Y$?
   - In logic, P ∨ F ≡ P. Correct law **1 mark**

   Explanation **1 mark**

   Law: Identity Law (Law of Identity)
3. Define adjacency matrix and explain its structure with an example.
   - A square matrix used to represent a finite graph.

     If there is an edge between vertices i and j, entry a_{ij}= 1; otherwise 0.



1 mark

Example: $\begin{matrix} \\ a \\ b \\ c \end{matrix} \begin{matrix} \begin{matrix} a & b & c \end{matrix} \\ \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \end{matrix}$. **1 mark**

4. What is a p-value?
   - The probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis is true. **2 marks**
5. State whether tautology, contradiction, and contingency are mutually exclusive. Explain shortly.
   - Yes, they are mutually exclusive. **1 mark**

     A tautology is always true, a contradiction is always false, and a contingency is sometimes true and sometimes false. **1 mark**
6. Compare distributive law in sets with distributive law in arithmetic.
   - Set theory: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ **1 mark**

     Arithmetic: $a \times (b + c) = a \times b + a \times c$

   Comparison: Both show how one operation distributes over another.**1 mark**
7. What is logical equivalence?
   - Two statements are logically equivalent if they have the same truth value in every possible situation.

     Notation: $P \equiv Q$. **2 marks**

Part B

Short Essay Questions (5 marks each)

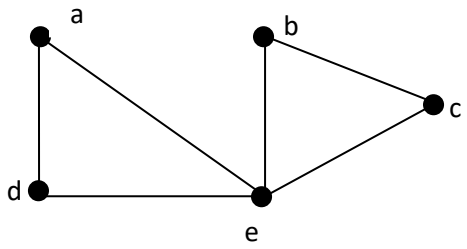
Answer any 6 questions

(Total = 6 × 5 = 30 marks)

8. Explain why a connected graph with n vertices and $(n-1)$ edges must be a tree.
   - A connected graph with n vertices and $n-1$ edges has no cycles ⇒ it is minimally connected ⇒ hence a tree.

   Concept: **2 marks**
   Explanation: **2 marks**
   Conclusion **1 mark**
9. Differentiate between bar chart, histogram, and pie chart.
   - Bar chart: discrete data; gaps between bars.
   - Histogram: continuous data; no gaps.
   - Pie chart: circular chart showing percentage share.

   Any 2 points each **3 marks**
   Diagram/clarity **2 marks**
10. Consider a graph with 5 vertices, 6 edges. Draw a possible undirected graph and label the vertices.



   Correct graph **3 marks**
   Labeling **2 marks**
11. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Compute $A \times I$ and $I \times A$ where I is the identity matrix.
   - $A \times I = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = A$; $I \times A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = A$.

   Multiplication 4 marks
   Result explanation 1 mark
12. State commutative laws of logic with examples.
   - For Conjunction (∨): $P \vee Q \equiv Q \vee P$
   - For Disjunction (∧): $P \wedge Q \equiv Q \wedge P$
   - Example with truth table or verbal justification.

   Laws **2 marks**
   Example **2 marks**
   Explanation **1 marks**
13. Write short notes on Hamiltonian cycle and give example graph which has it
   - A Hamiltonian cycle passes through every vertex exactly once and returns to start.
   - Example: square + diagonal (4 vertices).

   Definition **2 marks**
   Example **2 marks**
   Explanation **1 mark**

14. A coin is tossed 3 times. Find the probability distribution of the number of heads.
   - x = Number of heads.

$P(X=0)=1/8, P(X=1)=3/8, P(X=2)=3/8, P(X=3)=1/8.$

Listing outcomes **2 marks**

Probabilities **3 marks**

15. Construct truth table for $(p \wedge q) \wedge \neg q$.

p	q	$\sim q$	$p \wedge q$	$(p \wedge q) \wedge \sim q$
T	T	F	T	F
T	F	T	F	F
F	T	F	F	F
F	F	T	F	F

- The last column shows false (contradiction).

Correct table **3 marks**

Conclusion **1 mark**

Part B

Essay Questions (10 marks each)

Answer any 3 questions

(Total = 3 × 10 = 30 marks)

16. Solve by matrix method:

$x + y = 5$

$.2x - y = 1$

$3x + y = 7$

- Represent as $AX = B$ $A = \begin{pmatrix} 1 & 1 \\ 2 & -1 \\ 3 & 1 \end{pmatrix}$ $Ans: x = 2, y = 3.$

Matrix form **3 marks**

Steps/inverse **5 marks**

Final answer **2 marks**

17. Define a conditional statement with suitable examples. Explain its truth table clearly.
    - Definition: "If P, then Q" (P → Q). **2 marks**
    - Example: If it rains, then the ground is wet. **2 marks**
    - Truth table (4 rows). **4 marks**
    - Explanation of when false (P true, Q false) **2 marks**
18. Define connected graphs. Explain the difference between connected & disconnected graphs with diagrams.
    - Connected: every pair of vertices joined by a path.
    - Disconnected: at least one pair not connected.
    - Diagram: one of each type.

    Definition **2 marks**

    Explanation **4 marks**

    Diagram **4 marks**
19. Define and distinguish between Mean, Median and Mode. Explain with examples.
    - Mean: arithmetic average.
    - Median: middle value.
    - Mode: most frequent value.
    - Comparison + example dataset.

Definitions **3 marks**
Example **3 marks**
Comparison **4 marks**

20. What is meant by logic in mathematics? Why is it considered important in solving mathematical problems?
    - Logic: study of reasoning principles in mathematics.
    - Importance: ensures correctness of proofs, formulation of arguments, algorithm design, decision-making.

Definition **3 marks**
Explanation **3 marks**
Importance/examples **4 marks**